

ADAM v30: Complete Raspberry Pi Zero 2W Hardware Guide

This is your master schematic for the Raspberry Pi Zero 2W. All pins refer to the physical pin numbers (1-40) on the Pi's header block.

1. TFT Display (ILI9341 2.42")

We use Hardware SPI0 for high-speed screen drawing via `adam_tft.py`.

- **MOSI:** Pin 19 (GPIO 10)
- **SCLK / SCK:** Pin 23 (GPIO 11)
- **CS / CE:** Pin 24 (GPIO 8 / CEO)
- **DC / Data:** Pin 18 (GPIO 24)
- **RST / Reset:** Pin 22 (GPIO 25)
- **VCC:** Pin 17 (3.3V)
- **GND:** Pin 20 (GND)
- **LED / BLK:** Pin 1 (3.3V) — powers the backlight.

2. Pan Servo

- **Signal (Orange/Yellow):** Pin 32 (GPIO 12). This is a hardware PWM pin, which prevents the servo from shaking/jittering.
- **VCC & GND:** Connect directly to your external 5V buck converter. Do NOT use the Pi's 5V pin.

3. Stereo Microphones (2x INMP441)

Since you are using two microphones, they must share the I2S bus. You differentiate them by changing how their L/R (Left/Right Channel) pins are wired. This setup allows ADAM to calculate sound direction (DoA).

- **SCK / BCLK (Clock):** Pin 12 (GPIO 18) — **Connects to BOTH Mics**
- **WS / LRC (Word Select):** Pin 35 (GPIO 19) — **Connects to BOTH Mics**
- **SD / DIN (Data In):** Pin 38 (GPIO 20) — **Connects to BOTH Mics**
- **VDD:** 3.3V (Any 3.3V pin)
- **GND:** GND (Any ground pin)
- **L/R on MIC 1 (Left):** Wire to GND.
- **L/R on MIC 2 (Right):** Wire to VDD (3.3V).

4. Audio Amplifier (MAX98357A)

The amplifier *also* shares the exact same clock signals as the microphones on the I2S bus, but outputs data instead.

- **BCLK (Clock):** Pin 12 (GPIO 18) — **Shares with Mics**
- **LRC (Word Select):** Pin 35 (GPIO 19) — **Shares with Mics**
- **DIN (Data Out):** Pin 40 (GPIO 21) — **Unique to Amplifier**
- **VIN / Power:** Connect directly to your external 5V buck converter. Do NOT use the Pi's 5V pin!
- **GND:** Connect to shared ground.
- **Speaker terminals (+/-):** Connect to your 40mm 3W speaker.

Software Configuration

1. API Keys and IP Variables

In the same folder as `adam_main_wifi.py` and `adam_tft.py`, you must create a file literally named `.env` (with the leading dot). Place this inside:

```
GEMINI_API_KEY="AlzaSy...Your_Actual_Key_Here..."
ESP32_IP="192.168.1.xxx"
```

2. Raspberry Pi Config File

To turn on the SPI display and the I2S audio bus, you must edit the system boot configuration.

Open terminal: `sudo nano /boot/firmware/config.txt` (or `/boot/config.txt`)

Ensure these lines are uncommented or added to the bottom:

```
dtoverlay=spi=on
dtoverlay=i2s=on
dtoverlay=googlevoicehat-soundcard
```

Note: The `googlevoicehat-soundcard` overlay allows an I2S mic and I2S speaker to run simultaneously.

3. Audio Selection

1. Click the Volume/Speaker icon in the top right of the Raspberry Pi Desktop GUI.
2. Ensure **Google voiceHAT Soundcard** is checked for both **Output** and **Input**.

4. Code Execution

Run the system using:

```
python3 adam_main_wifi.py
```

(You do not need to run adam_tft.py separately; the main script imports and runs it automatically).